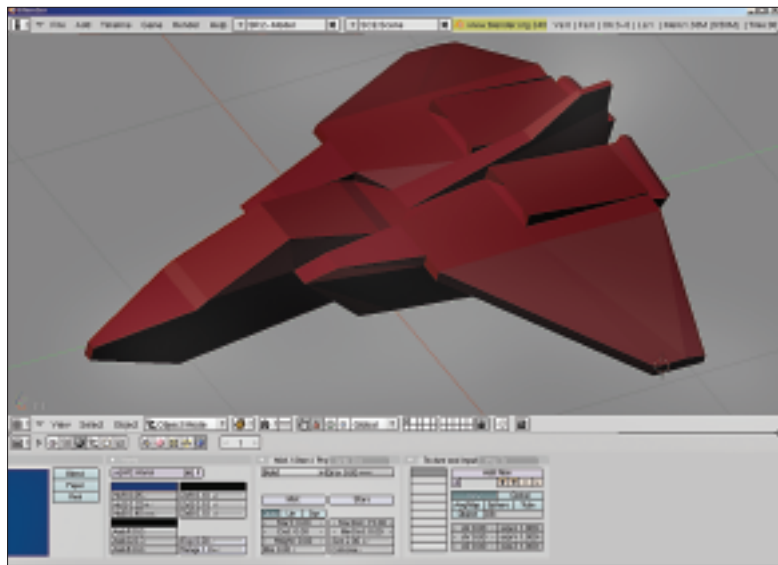




A 3D model in the open source program, Blender

typical of much open source software. The package is very well documented, with hundreds of pages of tutorials and documentation available for download and more information available on the Web site. The downloadable PDFs are very useful, but lacking some features of the software. Perhaps the Web site is kept more up-to-date.

The modelling uses standard meshes (primitives constructed from triangles), curves and surfaces using NURBS, and meta objects. The latter refers to a procedural method for defining surfaces; they are simply mathematical definitions. There are five basic types: ball, tube, plane, ellipsoid and cube. You can construct more complex objects by editing and combining these together, and then eventually convert to a standard mesh of triangles when satisfied with the basic model. Interestingly, meta objects interact with and influence each other when in close proximity, and there is a parameter called stiffness that determines just how strong these effects are. Basically, the surfaces that are close to each other move towards each other — they bulge out — and then combine in a fluid and uncomplicated manner (none of the artefacts than can develop when Boolean operations are performed on meshes). These effects can also be negative, where one surface pushes another away (without them actually touching).

This adds yet another type of control which may be suitable for certain types of model where polygon meshes or NURBS are not. Moving objects close to each other, the connections between them appear in real-time, and so this is a very fast method for developing certain combined effects — with Boolean operations you may have to repeat a process several times, undoing it each time,

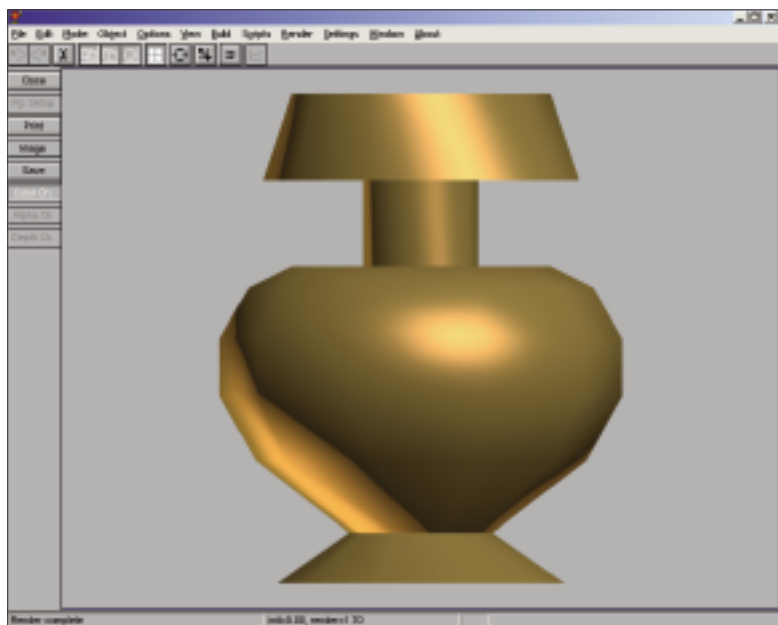
before you get it right. 3ds MAX users might be interested to know that files created in Blender can be exported to 3ds format and opened in MAX. The couple we have tried worked successfully, and so this makes this functionality available to MAX users. Some effects can be created using this method that would be quite difficult using MAX.

As far as lighting is concerned, Blender includes both ray tracing and radiosity for advanced lighting techniques.

Anim8tor

Anim8tor (www.anim8or.com) is a small free modelling and animation program (less than 2 MB!), with surprisingly good functionality, although far less than programs such as MAX and Blender. All you need do is run it from the desktop — there is no need for installation, or the installation of other modules as with Blender (that needs the Microsoft C++ runtime and Python). The current version number is 0.95, so clearly this software is still under development.

It is certainly very suitable for beginners, or those with not quite so demanding requirements for their models and animation. It would, for example, be very suitable for titles and simple animations for non-broadcast video productions, or for inclusion with business presentations. Documentation is provided on the Web site, and is also available for download. The downloadable manual is available in several languages other than English (Czech, Italian, Chinese, French, Portugese and Spanish) but it seems that only the English version matches the current progress in the software.



A simple model rendered within Anim8tor

There are also several tutorials on the Web site, some of which contain downloadable versions of the simple models that they describe. The site suggests a particular one (modelling an eggplant) as a good basis for learning the basics of modelling with a mesh, and it lives up to the claim quite well. A very good place to start.

Currently, there is no support for NURBS or other more sophisticated surface modelling techniques — this is strictly mesh-based modelling, but with the basic primitives such as spheres, cubes and so on.

According to the author, the most work needs to be done on the renderer, but the most important features are present for creating materials: ambient, diffuse and specular texture mapping, plus also emissive (when the surface itself radiates light), transparency and bumps. Considering that the software also supports some level of ray tracing, it is surprising that so much can be packed into 2 MB.

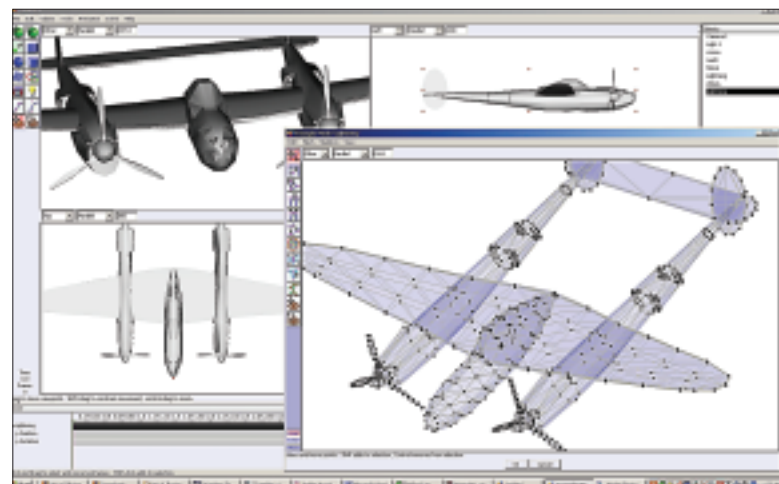
For compatibility with other software, Anim8tor can import the older 3ds formats from Autodesk 3D Studio, and also files from Lightwave and Wavefront. It can also export to 3D Studio and Wavefront formats.

Art Of Illusion

Available for Windows, Mac OS X, Linux and Unix, the open source Art of Illusion (www.artofillusion.org) falls between Blender and Anim8tor in complexity. Surprisingly, it is written in Java, and is available free of charge under the General Public License. There is extensive documentation available on the Web site, that can also be downloaded, but unfortunately this is not in a straightforward PDF format, but as a large jumble of HTML files. This makes it difficult to search through, and also difficult to find your way around, until you find the contents page and realise that it is the main navigation point.

Also available are good step-by-step tutorials, which can also be downloaded. These are again collections of HTML files, but highlight the key areas quite successfully: basic modelling, working with meshes, Boolean operations, procedural textures, and so forth.

This software enables the usual mesh-based modelling, with cube, sphere and cylinder primitives. Not an impressive list for such otherwise sophisticated software, but it also includes useful editing techniques, spline-based meshes and Boolean functionality; no NURBS, however. Another interesting modelling technique that is provided is that of the skin. Here, you define a series of curves or other shapes — rather like a skeletal structure — and the software creates a triangle map skin that fits around these.



Art of Illusion. Here, a model is to be edited at a vertex and triangle level. An extra editing window has been opened by the software

Interestingly, the software opens up a second window when you want to go in and edit a model at the vertex and triangle level. When you start the program, it opens up a standard set of four modelling windows. These can be set easily as views from the right, left, front, and so on, and when you want to switch to a single view for closer work, there is a simple hotkey that toggles between the two views. This seems even easier than the MAX icon method, even to this nearly two decade 3D Studio veteran. Some thought seems to have gone into the general usability of this product.

Art of Illusion supports the usual texture maps, including reflection, transparency and bump mapping. It also uses procedural mapping, including 3D procedures, in which objects are affected perpendicularly to the surface. In the simplest terms, this can be considered to provide more sophisticated techniques along the lines of bump mapping — but much more powerful.

It also supports advanced lighting techniques: ray tracing and global illumination. Strictly speaking, global illumination is usually considered to be a generic term for all such methods such as ray tracing and radiosity, but in this context, the authors are referring to a technique similar to radiosity.

Unfortunately, you can only import files from Wavefront — so you will need Anim8tor to convert your 3D Studio files in order to import them, just as we have done with the illustrations used here.

We have included here three typical examples worthy of attention, across the range of functionality, but there are other free tools available on the internet, and it is worth searching to see what else might suit your purposes — OpenFX, for example, is certainly worth a look. ■

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The gaming images in this article were provided to *Digit* by Kerry Fraser-Robinson, founder of Red Bedlam Ltd (www.redbedlam.com), developers of the UK's first true 3D MMORPG, and one of the leading developers of virtual world middle-ware